extension as a foundation for his equation and for his struggle to understand gravitational anomalies (Chapter 25).

If this speculative extension is correct, then time behaves fundamentally the same in the bulk as in our brane. In particular, objects and signals in the bulk, like those in our brane, can only move in one direction through locally measured time (local bulk time): toward the future. They cannot move backward, locally. If backward time travel is possible in the bulk, it can be achieved only by journeying out through the bulk's space and returning before the journey started while always moving forward in local bulk time. This is a bulk analog of the round trip in Figure 30.3.

Messaging Murph: My Physicist's Interpretation



This description of time underlies my physicist's interpretation of Cooper's messaging Murph.

Recall that the tesseract is an object whose faces have three space dimensions and interior has four. The interior is part of the bulk. Everything we see in the movie's tesseract scenes lies in the faces: Cooper, Murph, Murph's bedroom, the bedroom's extrusions, the world tubes of the book and watch—all lie in tesseract faces. We never see the tesseract's bulk interior. We can't see it, since light can't travel through four space dimensions, only three. However, gravity can do so.

In my interpretation, when Cooper sees a book in Murph's bedroom, he does so via a light ray that travels in faces of the tesseract (for example, the red dashed ray in Figure 30.5). And when he pushes on a book's world tube, or on the world tube of the watch's second hand, he generates a gravitational signal (a gravitational wave in the bulk) that spirals into and through the tesseract's bulk interior, along the violet